

HOMEWORK 7

INTRODUCTION TO COMPLEX ANALYSIS

(MATH 4230-40), SPRING 2022

SECTION 2.4 - CAUCHY'S INTEGRAL FORMULA

1. Evaluate the following integrals:

(a) $\int_{\gamma} \frac{z^2 - 1}{z^2 + 1} dz$, where γ is a circle of radius 2, centered at 0.

Let γ_1 be the circle of radius 1/2 centered at i and let γ_2 be the circle of radius 1/2 centered at $-i$. By an extension of the deformation theorem (i.e., that given in Example 2.2.9), it follows that

$$\int_{\gamma} \frac{z^2 - 1}{z^2 + 1} dz = \int_{\gamma_1} \frac{z^2 - 1}{z^2 + 1} dz + \int_{\gamma_2} \frac{z^2 - 1}{z^2 + 1} dz.$$

Let $f_1(z) = \frac{z^2 - 1}{z + i}$ and let $f_2(z) = \frac{z^2 - 1}{z - i}$. Then, we have

$$\int_{\gamma_1} \frac{z^2 - 1}{z^2 + 1} dz = \int_{\gamma_1} \frac{f_1(z)}{z - i} dz \quad \text{and} \quad \int_{\gamma_2} \frac{z^2 - 1}{z^2 + 1} dz = \int_{\gamma_2} \frac{f_2(z)}{z + i} dz.$$

Since f_1 is analytic on $\mathbb{C} \setminus \{-i\}$ and γ_1 is a closed curve homotopic to a point in this region, Cauchy's integral formula (i.e. Theorem 2.4.6) yields that

$$f_1(i)I(\gamma_1; i) = \frac{1}{2\pi i} \int_{\gamma_1} \frac{f_1(z)}{z - i} dz.$$

That is, $\int_{\gamma_1} \frac{f_1(z)}{z - i} dz = -2\pi$. Similarly, since f_2 is analytic on $\mathbb{C} \setminus \{i\}$ and γ_2 is a closed curve homotopic to a point in this region, another application Cauchy's integral formula yields that

$$f_2(-i)I(\gamma_2; -i) = \frac{1}{2\pi i} \int_{\gamma_2} \frac{f_2(z)}{z + i} dz.$$

That is, $\int_{\gamma_2} \frac{f_2(z)}{z + i} dz = 2\pi$. Therefore, $\int_{\gamma} \frac{z^2 - 1}{z^2 + 1} dz = -2\pi + 2\pi = 0$.

(b) $\int_{\gamma} \frac{\sin e^z}{z} dz$, where γ is the unit circle.

Let $f(z) = \sin e^z$. Then, $\int_{\gamma} \frac{\sin e^z}{z} dz = \int_{\gamma} \frac{f(z)}{z - 0} dz$. Since f is analytic on $\mathbb{C}$ and γ is homotopic to a point in $\mathbb{C}$, Cauchy's integral formula (i.e. Theorem 2.4.4) yields that

$$f(0)I(\gamma; 0) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - 0} dz.$$

Rearranging and noting that $I(\gamma; 0) = 1$, this says that

$$\int_{\gamma} \frac{\sin e^z}{z} dz = 2\pi i \sin e^0 = 2\pi i \sin(1).$$

2. Let f be entire and assume that $|f(z)| \leq M|z|^n$ for large $|z|$, for a constant M , and some non-negative integer n . Show that f is a polynomial of degree less than or equal to n .

We will show that f is a polynomial of degree less than or equal to n using induction. In case you are not familiar with this technique, this means that we will show that our desired claim is true for the 'base case' of $n = 0$, then show that making an 'inductive assumption' that our claim is true for *some* value of n *also* implies that it is true for $n + 1$ as well. Naturally, applying this fact to our base case of $n = 0$ will then yield the case of $n = 1$, which will then yield the case of $n = 2$, and so on.

Base Case: To begin, we start with the case of $n = 0$ and suppose $|f(z)| \leq M|z|^0 = M$ for large $|z|$ and some constant M . Then, by Liouville's theorem, f is constant—i.e., a degree $n = 0$ polynomial. So, the assertion holds in the case of $n = 0$.

Inductive Step: Now, suppose that our desired assertion is true for some fixed value of n . Let f be an entire function and assume that $|f(z)| \leq M|z|^{n+1}$ for large $|z|$ and some constant M . Define

$$F(z) = \begin{cases} \frac{f(z) - f(0)}{z} & z \neq 0 \\ f'(0) & z = 0 \end{cases}$$

We first show that F is continuous on $\mathbb{C}$. From the continuity of f , the continuity of F is immediate everywhere except potentially at $z = 0$. So, note that having f differentiable at $z = 0$ implies that $\lim_{z \rightarrow 0} \frac{f(z) - f(0)}{z - 0} = f'(0)$, yielding continuity at $z = 0$.

Now, we show that F is analytic. Since f is entire, F is certainly analytic on $\mathbb{C} \setminus \{0\}$. By Corollary 2.4.11, F is in fact analytic on $\mathbb{C}$, and therefore entire as well.

Let us now assume now that $f(0) = 0$, since we could replace f with $\tilde{f}(z) = f(z) - f(0)$ were this not the case (that is, if we can show that $\tilde{f}(z) = p(z)$ for some polynomial p , then since $f(z) = \tilde{f}(z) + f(0)$, it would follow that $f(z) = p(z) + f(0)$ —which is also a polynomial). So, notice that for large $|z|$, we have

$$|F(z)| = \left| \frac{f(z) - f(0)}{z} \right| = \frac{|f(z) - f(0)|}{|z|} = \frac{|f(z)|}{|z|} \leq \frac{M|z|^{n+1}}{|z|} = M|z|^n.$$

So, $F(z)$ is a polynomial of degree less than or equal to n by our inductive assumption; i.e. we have $F(z) = a_0 + a_1 z + \dots + a_n z^n$ for some $a_0, \dots, a_n \in \mathbb{C}$. Since $zF(z) + f(0) = f(z)$, we then have

$$f(z) = f(0) + a_0 z + a_1 z^2 + \dots + a_n z^{n+1}.$$

As this is a polynomial of degree $n + 1$, our desired result follows by induction.

3. Consider the function $f(z) = \frac{1}{z^2}$.

- (a) Notice that $\int_{\gamma} f(z)dz = 0$ for all closed contours γ not passing through the origin, but is not analytic at $z = 0$. Does this statement contradict Morera's Theorem?

No, this does not contradict Morera's theorem. Morera's theorem is as follows:

Morera's Theorem: Let f be continuous on a region A and suppose that $\int_{\gamma} f = 0$ for every closed curve in A . Then, f is analytic on A and $f = F'$ for some analytic function F on A .

Notice, Morera's theorem requires both that the function f is continuous and that $\int_{\gamma} f = 0$ for all closed contours γ in the region A on which f is analytic. For any region A containing 0, then since $f(z)$ is not continuous at 0, the conditions of Morera's theorem are not satisfied and, correspondingly, its conclusion does not apply to A . For regions A not containing 0, the conditions are then satisfied and Morera's theorem does indeed apply.

- (b) Notice that f is bounded as $|z| \rightarrow \infty$, but is not a constant. Does this statement contradict Liouville's theorem?

No, this does not contradict Liouville's theorem. Liouville's theorem is as follows:

Liouville's Theorem: If f is entire and there is a constant M such that $|f(z)| \leq M$ for all $z \in \mathbb{C}$, then f is constant.

Notice, Liouville's theorem requires that f be entire; i.e., analytic on $\mathbb{C}$. Since $f(z) = \frac{1}{z^2}$ is not entire (f is not even defined on all of $\mathbb{C}$), the conditions of Liouville's theorem are not satisfied and, correspondingly, its conclusion does not apply.